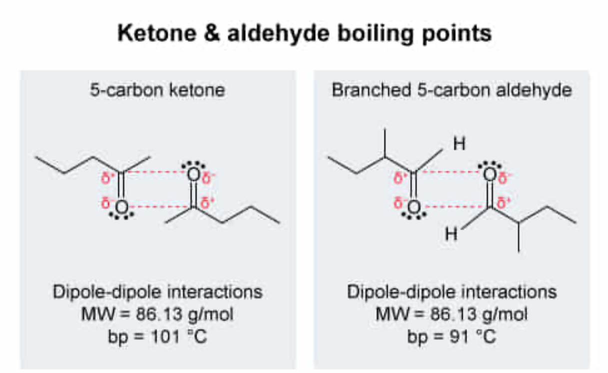
A linear five-carbon ketone has a boiling point most similar to:

- A. a five-carbon alkane. [9%]
- B. a five-carbon carboxylic acid. [8%]
- C. a branched aldehyde with the same molecular weight. [75%]
- D. an alcohol with a higher molecular weight. [5%]

Correct
75% answered correctly

Explanation:



Ketones and aldehydes experience the same intermolecular forces. Those with the same molecular weight have similar boiling points.

The **boiling point** of a substance depends on the intermolecular forces holding the molecules together. **Strong intermolecular forces** require more energy to break, causing **higher boiling points**. Molecules with functional groups that participate in strong bonds experience stronger intermolecular forces and have higher boiling points than molecules with similar molecular weights (MWs) that participate only in weak bonds. Molecules with a greater surface area (higher MW or less branching) experience more intermolecular forces and have **higher boiling points** than similar molecules with smaller surface areas.

The carbonyls of ketones and aldehydes have a partial positive charge on the carbon atom and a partial negative charge on the oxygen atom, forming a dipole moment and allowing dipole-dipole interactions to occur. Branching causes a relatively small boiling point difference in compounds with similar functional groups and MWs. Therefore, a five-carbon ketone and a branched aldehyde of the same MW have similar boiling points.

(Choice A) Alkanes experience only London forces, whereas ketones experience London forces *and* dipole-dipole interactions. Therefore, ketones participate in stronger intermolecular forces and have significantly higher boiling points than alkanes of the same carbon chain length.

(Choice B) Carboxylic acids experience the same interactions as ketones and aldehydes, but also contain hydrogen bond donors and acceptors, and can hydrogen bond with themselves. Therefore, carboxylic acids participate in stronger intermolecular forces and have significantly higher boiling points than ketones of the same carbon chain length.

(Choice D) Alcohols can act as hydrogen bond donors and acceptors and therefore can hydrogen bond with themselves. Hydrogen bonds are stronger than dipole-dipole interactions and give alcohols significantly higher boiling points than ketones or aldehydes of the same MW. An alcohol of *higher* MW would have an even higher boiling point.

Educational objective:
Boiling point depends on intermolecular forces, surface area, and molecular weight (MW). Strong intermolecular forces and greater surface area (less branching and higher MW) have higher boiling points compared to molecules with weaker intermolecular forces and a smaller surface area. Ketones and aldehydes of the same MW have similar boiling points because they both have dipole-dipole interactions.

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Functional Groups and Their Reactions